Lokesh Meruva

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Summary

AI/ML engineer experienced in building **agentic AI systems**, **multi-agent pipelines**, and **RAG architectures** using LangGraph, vector databases, and Python – from voice-based multi-agent assistants to MCP-based tool orchestration servers. Strong grounding in prompt engineering, Azure cloud deployment, and production APIs with FastAPI.

Education

KL University — Vijayawada, India
B.Tech in Computer Science & Engineering (AI/ML Specialization); CGPA: 8.72/10 — *Expected June 2026*

Technical Skills

Languages: Python, SQL
Agentic AI & Orchestration: LangChain, LangGraph, Multi-Agent Systems, Model Context Protocol (MCP), FastMCP, Prompt Engineering, Agent Reasoning Patterns
RAG & Retrieval: Retrieval-Augmented Generation (RAG), RAGAS Evaluation, Vector Databases (ChromaDB), Embeddings
Document AI & OCR: OCR Pipelines, PDF/Document Data Extraction, Speech-to-Text (faster-whisper)
APIs & Deployment: FastAPI, REST APIs, Docker, Microsoft Azure, Streamlit, Git/GitHub
Databases: SQLite, MongoDB, Google Sheets API
Concepts: Deep Learning, NLP, Computer Vision, Model Evaluation & Latency Optimization

Experience

AICTE – Microsoft Azure AI Internship — Remote
AI Intern (Edunet Foundation) — *Jan 2025 – Mar 2025*

- Built and deployed AI/ML models on **Microsoft Azure**, gaining hands-on experience with cloud-based model training, deployment, and evaluation workflows.
- Applied cloud AI services to real-world use cases, strengthening skills directly applicable to deploying agentic AI pipelines on Azure.

Projects

VEDA – Multi-Agent Voice Assistant | *LangGraph, ChromaDB, Python, faster-whisper*

- Designed a **multi-agent orchestration pipeline** using **LangGraph** where specialized agents reason, retrieve context via **ChromaDB**, and act autonomously on voice commands.
- Built a **CPU-only**, low-latency speech pipeline with **faster-whisper** supporting Telugu/English code-switching and custom wake-word detection.

RAG-Optimizer – RAG Benchmarking Studio | *Python, RAGAS, Streamlit*

- Built an 18-configuration **RAG pipeline** benchmarking tool evaluating chunking strategies, retrievers, and embeddings for structured and unstructured data using **RAGAS** metrics.
- Surfaced accuracy, latency, and reliability comparisons via an interactive Streamlit leaderboard to guide model/config selection.

Multi-Tool MCP Server – Agentic Tool Orchestration | *FastMCP, Python, FastAPI*

- Developed a **FastMCP**-based server exposing tools that let an AI agent read/write **Google Sheets**, query **SQLite**, and extract data from **PDFs** via **FastAPI** endpoints – enabling agentic, multi-tool workflows.

Publications & Certifications

- **CNN-Based Traffic Severity Classification** – Accepted and Presented at **BIDA 2026**; Springer LNCS publication pending.
- **Certifications:**Salesforce Certified AI Associate · MongoDB Associate DBA · Aviatrix Certified Engineer (ACE)